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A CONDITIONAL SPECTRAL DETERMINANT FRAMEWORK FOR THE RIEMANN XI FUNCTION

Main result: a logarithmic δ-barrier Schrödinger operator is self-adjoint, with positive simple spectrum, trace-class inverse and an exact transfer formula; under synthetic Axiom (S) its Fredholm determinant equals ξ(s)/ξ(½), which implies RH and the simplicity of all nontrivial zeros. Axiom (S) is assumed, not proved: this is a conditional framework, not a proof of RH.

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Abstract

We formulate a synthetic spectral correspondence hypothesis for a positive Schrödinger operator on the half-line with logarithmically spaced delta interactions. The operator is defined independently of the Riemann zeta function. Its self-adjointness, confinement, simple spectrum and trace-class inverse follow from the stated barrier data. An exact transfer formula expresses its resolvent trace through an initial-value problem.

A separate synthetic axiom identifies this trace with the logarithmic derivative of the completed zeta function. Assuming that axiom, we prove an entire Fredholm determinant identity, a correspondence covering every nontrivial zeta zero, the Riemann hypothesis and simplicity of those zeros. The existence of parameters satisfying the axiom is not proved. This paper is a conditional framework, not an unconditional proof of RH. A separate zero-indexed reflection construction illustrates why signed cancellation does not imply that a positive obstruction vanishes.

1 Model and conventions

Work in units $\hbar = 2m = 1$. Fix positive parameters x_0 and g_0, and set

$$x_k = x_0 \log k, \qquad g_k = g_0 k^{-1/2}, \qquad k \geq 2.$$

The index starts at 2 so that no interaction is placed at the Dirichlet boundary. On $L^2(0,\infty)$, define the nonnegative quadratic form

$$q[f] = \int_0^\infty |f'(x)|^2 \, dx + \sum_{k=2}^{\infty} g_k |f(x_k)|^2,$$

$$\mathcal{Q} = \left\{ f \in H_0^1(0,\infty) : \sum_{k=2}^{\infty} g_k |f(x_k)|^2 < \infty \right\}.$$

Point values refer to the continuous Sobolev representative. Let H denote the operator associated with these parameters. All zero correspondences are equalities of multisets.

2 Self adjointness and confinement

Proposition 1. For every $x_0, g_0 > 0$, the form q is densely defined, closed and nonnegative. Its associated operator H is self-adjoint, has compact resolvent and has strictly positive simple eigenvalues.

Proof of the form assertion. Compactly supported smooth functions belong to the form domain because the interaction set is locally finite. A Cauchy sequence in the form norm converges in H_0^1, while its weighted point-value vectors converge in ℓ^2. Continuity of each point evaluation identifies the limits. This proves closedness; the representation theorem supplies H. For related form methods for δ-interactions, see [1].

Put $I_1 = (0, x_2)$ and $I_k = (x_k, x_{k+1})$ for $k \geq 2$. An explicit operator domain consists of those f in Q whose restrictions belong to $H^2(I_k)$, whose piecewise second derivative is square integrable, and which satisfy

$$\sum_{k \geq 1} \|f''\|^2_{L^2(I_k)} < \infty, \qquad f'(x_k +) - f'(x_k -) = g_k f(x_k).$$

Continuity and f(0) = 0 are encoded in Q; Hf = −f″ between barriers. Integration by parts gives this domain. Cutoff approximation in the form norm justifies the identification without an additional boundary condition at infinity.

For confinement, let $d_k = x_{k+1} - x_k$. The fundamental theorem of calculus gives

$$\int_{x_k}^{x_{k+1}} |f|^2 \leq 2d_k |f(x_k)|^2 + 2d_k^2 \int_{x_k}^{x_{k+1}} |f'|^2.$$

Since $d_k \leq x_0/k$ and $d_k/g_k \leq x_0/(g_0\sqrt{k})$, summation gives

$$\int_{x_K}^{\infty} |f|^2 \leq \left(\frac{2x_0}{g_0\sqrt{K}} + \frac{2x_0^2}{K^2} \right) q[f].$$

Bounded form-norm sets therefore have uniformly vanishing tails. Rellich compactness on finite intervals proves that Q embeds compactly into L^2, hence H has compact resolvent.

If Hf = 0, then q[f] = 0 forces f′ = 0, hence f = 0. Every eigenspace is at most one dimensional: f(0) = 0 and the scalar f′(0+) determine the solution uniquely, and the jump conditions propagate it through all barriers. Thus

$$0 < E_1 < E_2 < \cdots, \qquad E_n \longrightarrow \infty. \qquad \blacksquare$$

3 Trace class inverse and transfer formula

Proposition 2. The inverse H^{-1} is trace class. To see this, use the tail bound in Section 2, writing its coefficient as A_K. For large E choose $K = O(E^2)$, with a sufficiently large constant, so that $A_K^{-1} > E$. Split at x_K, allow independent boundary values there, and assign the barrier at x_K to the tail. This enlarges the form domain. The tail has no eigenvalues at or below E; discarding the remaining barriers on the finite interval gives the free Dirichlet–Neumann comparison. Therefore

$$N_H(E) \le 1 + \frac{x_K\sqrt{E}}{\pi} = O(\sqrt{E}\log E), \qquad \sum_n E_n^{-1} < \infty.$$

For $\kappa \ge 0$ define u_κ as the initial-value solution of the differential expression $(H + \kappa^2)u_\kappa = 0$, with $u_\kappa(0) = 0$ and $u_\kappa'(0) = 1$. This is a generalized solution, not a vector asserted to belong to D(H) or L^2. It is continuous and has the same derivative jumps as the operator domain.

For the state vector $(u_\kappa, u_\kappa')^T$, free propagation through a length d and passage through the kth barrier use, respectively,

$$u(x+d) = u(x)\cosh(\kappa d) + u'(x)\sinh(\kappa d)/\kappa$$
$$u'(x+d) = \kappa u(x)\sinh(\kappa d) + u'(x)\cosh(\kappa d)$$
$$u(x_k+) = u(x_k-), \quad u'(x_k+) = u'(x_k-) + g_k u(x_k)$$

At $\kappa = 0$ use the continuous limit. Starting with $(0,1)^T$, alternate free propagation and barrier updates to construct u_κ independently of zeta. For L = x_N, let H_L have Dirichlet conditions at both endpoints; the barrier at L contributes zero. The one-dimensional Gelfand–Yaglom ratio formula [2, Section 3] takes the following form:

$$\frac{u_\kappa(L)}{u_0(L)} = \det(I + \kappa^2 H_L^{-1}), \qquad \partial_\kappa \log u_\kappa(L) = 2\kappa \mathrm{Tr}(H_L + \kappa^2 I)^{-1}.$$

For the finitely many interior δ-interactions, the ratio formula can also be verified directly. Write $t = \kappa^2$ and let y_t be the left initial-value solution and z_t the right solution with $z_t(L) = 0$ and $z_t'(L) = -1$. Differentiating the equation in t and integrating the Wronskian identity gives $\partial_t y_t(L) = \int_0^L y_t(x) z_t(x)\, dx$. The Wronskian is continuous at each barrier because both solutions obey the same jump condition. The Green kernel has diagonal $y_t(x)z_t(x)/y_t(L)$, so $\partial_t \log y_t(L) = \mathrm{Tr}(H_L + tI)^{-1}$. Integration from t = 0 proves the determinant ratio above, normalized to 1 at zero.

Dirichlet exhaustion gives $E_n(L) \downarrow E_n$. Domination by the summable sequence E_n^{-1} justifies passage to the trace limit. In particular, for $s > 1$ and $\kappa = (s - 1/2)/c$,

$$\frac{1}{c}\lim_{N\to\infty} \partial_\kappa \log u_\kappa(x_N) = 2(s-\tfrac{1}{2})\mathrm{Tr}(c^2 H + (s-\tfrac{1}{2})^2 I)^{-1}.$$

The derivative is taken before the limit. For real $\kappa \ge 0$ the initial-value solution is positive at every $x > 0$, so the logarithm in this formula is well-defined.

4 Synthetic axiom and conditional theorem

Let Λ be the von Mangoldt function: $\Lambda(p^r) = \log p$ for primes p and integers $r \geq 1$, and $\Lambda(m) = 0$ otherwise. Define

$$A(s) = \tfrac{1}{s} + \tfrac{1}{s-1} - \tfrac{1}{2}\log\pi + \tfrac{1}{2}\tfrac{\Gamma'(s/2)}{\Gamma(s/2)}.$$

Synthetic axiom (S)

There exist $x_0, g_0, c > 0$ such that, for every real $s > 1$,

$$\tfrac{1}{c}\lim_{N\to\infty} \partial_\kappa \log u_\kappa(x_0 \log N)\Big|_{\kappa = (s-1/2)/c} = A(s) - \sum_{m=2}^{\infty} \tfrac{\Lambda(m)}{m^s}.$$

The Dirichlet series converges absolutely in this range. Axiom (S) supplies the exact arithmetic matching. It is assumed, not derived from the logarithmic barrier locations or the strengths $k^{-1/2}$. No parameter triple satisfying (S) is exhibited here.

Use the completed zeta function [3]

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

It is entire; its zeros, with multiplicity, are exactly the nontrivial zeta zeros. Its value at $1/2$ is nonzero. For $s > 1$, the right side of S equals $\xi'(s)/\xi(s)$.

Conditional theorem

Assuming axiom (S), the following identity holds on the whole complex plane:

$$\xi(s) = \xi(\tfrac{1}{2})\det\Big(I + \tfrac{(s-\frac{1}{2})^2}{c^2} H^{-1}\Big).$$

Consequently, as multisets,

$$\{\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\} = \{\tfrac{1}{2} \pm ic\sqrt{E_n} : n \geq 1\}.$$

Thus axiom S implies RH and simplicity of every nontrivial zero. The quantification over all nontrivial zeros is essential: an identification restricted to zeros already on the critical line would not imply RH.

5 Proof and mathematical status

Proof of the conditional theorem. Since H^{-1} is trace class, the Fredholm determinant is the locally uniformly convergent entire product

$$D(s) = \prod_{n=1}^{\infty} \left(1 + \frac{(s-\frac{1}{2})^2}{c^2 E_n}\right), \qquad D(\tfrac{1}{2}) = 1.$$

Away from its zeros, termwise logarithmic differentiation gives

$$\frac{D'(s)}{D(s)} = 2(s-\tfrac{1}{2}) \sum_{n=1}^{\infty} \frac{1}{c^2 E_n + (s-\frac{1}{2})^2}.$$

By the independently derived transfer formula and axiom (S), $D'/D = \xi'/\xi$ for real $s > 1$. Both functions are nonzero there, so ξ/D is a constant a on that interval. The entire function $\xi - aD$ vanishes on an interval and hence vanishes identically. Evaluation at $s = 1/2$ gives $a = \xi(1/2)$, proving (D).

The convergent determinant product vanishes exactly when one factor vanishes. Since $E_n > 0$, these points are $1/2 \pm ic\sqrt{E_n}$. Each factor has a simple zero at each of its two roots; distinct eigenvalues give distinct roots. The zero multisets of ξ and D therefore agree, proving both RH and simplicity under S. □

Scope of the result

The operator analysis and transfer formula are unconditional for this family. The spectral–arithmetic equality is a separate axiom. Differentiating identity (D) recovers (S) on $s > 1$; thus (S) and (D) are equivalent within the established operator framework. The axiom packages the unresolved correspondence rather than deriving it. RH alone would not specify this operator or imply simplicity of all zeros.

A future proof must derive the arithmetic side directly from the barrier dynamics, including the selection of prime powers, the log p weights and the gamma contribution. Finite coefficient fits, numerical zero matches and leading counting asymptotics cannot replace the exact identity.

6 A reflection construction and its limitation

This section concerns a separate zero-indexed construction. It is not a derived component of the delta-barrier operator H, and no identification between the two Hilbert spaces or their operators is assumed. Let Z be the multiset of all nontrivial zeta zeros, with a separate index for each occurrence of a repeated zero. Define

$$\mathcal{K} = \ell^2(Z), \qquad Ce_\rho = (1/2 - \mathrm{Re}\rho)e_\rho, \qquad P = C^2 \geq 0.$$

Since every nontrivial zero satisfies $0 < \mathrm{Re}\,\rho < 1$ [3], C is an everywhere-defined bounded self-adjoint diagonal operator with norm at most $1/2$. Thus P is bounded and positive, and

$$Pe_\rho = (1/2 - \mathrm{Re}\rho)^2 e_\rho, \qquad P = 0 \;\Leftrightarrow\; \mathrm{RH}.$$

This equivalence encodes RH in an operator defined from the zeros themselves. It does not establish that P vanishes.

Reflection lemma

On $\mathcal{K} \oplus \mathcal{K}$ define the following bounded operators, where I is the identity on $\mathcal{K}$. The symbol S in this section denotes the swap operator; it is distinct from synthetic axiom (S).

$$R = \begin{pmatrix} P & 0 \\ 0 & -P \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}.$$

Then R and S are self-adjoint, S is an involution, and it reverses the sign of R:

$$R^* = R, \qquad S^* = S, \qquad S^2 = I, \qquad SRS = -R.$$

For every $f \in \mathcal{K}$, set $v = (f,f)/\sqrt{2}$. This vector has norm $\|f\|$ and is a unit vector when $\|f\| = 1$. Its quadratic form is

$$v = \frac{(f,f)}{\sqrt{2}}, \qquad \langle Rv, v\rangle = \frac{1}{2}\langle Pf, f\rangle - \frac{1}{2}\langle Pf, f\rangle = 0.$$

Proof. Since $P = P^*$, taking adjoints of the two block matrices gives $R^* = R$ and $S^* = S$. Multiplication gives $S^2 = I$ and

$$SRS = \begin{pmatrix} -P & 0 \\ 0 & P \end{pmatrix} = -R, \qquad Rv = \frac{(Pf, -Pf)}{\sqrt{2}}.$$

Expanding the inner product of Rv with v gives the two equal terms with opposite signs displayed above. This proves the lemma. □

Why cancellation is not vanishing

The paired quadratic form is zero for every bounded positive P, including nonzero P. For a hypothetical displacement δ = 0.1, the squared displacement is 0.01 and the corresponding paired block is

$$R_\delta = \begin{pmatrix} 0.01 & 0 \\ 0 & -0.01 \end{pmatrix}, \qquad w = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \langle R_\delta w, w \rangle = 0.$$

The paired expectation is zero, but Rδ is not the zero matrix: $R\delta w = (0.01, -0.01)^{\mathrm{T}}/\sqrt{2} \neq 0$. This is an algebraic example, not a claim that a zeta zero with that displacement exists. Reflection cancels signed contributions without forcing the original squared displacement to vanish.

Positivity would allow vanishing of a positive quadratic form to imply annihilation of that vector. Here R is indefinite whenever P ≠ 0, so that inference is unavailable. Moreover, the cancellation has only been asserted on the diagonal subspace of pairs (f,f). In contrast, squaring R gives

$$R^2 = \begin{pmatrix} P^2 & 0 \\ 0 & P^2 \end{pmatrix}, \qquad \langle R^2 v, v \rangle = \langle P^2 f, f \rangle = \| Pf \|^2.$$

This expression vanishes for every f exactly when P = 0. The unresolved RH condition therefore survives the reflection; a vanishing signed expectation alone supplies no such conclusion.

The reflection construction is an exact algebraic identity. It does not establish that the displacements vanish, does not prove RH, and does not discharge synthetic axiom S. Its purpose is to distinguish cancellation in an enlarged system from vanishing of the underlying positive obstruction.

7 Conclusion

The delta-barrier analysis establishes a self-adjoint operator with compact resolvent, positive simple spectrum, a trace-class inverse and an independent transfer formula. Under axiom (S), the determinant identity covers all nontrivial zeta zeros and implies RH and simplicity. No parameter triple satisfying (S) has been exhibited, and no unconditional proof of RH or zero simplicity is obtained. The separate reflection lemma does not supply the missing spectral–arithmetic correspondence.

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